

A Project Abstract

on

**ROUTE2CAMP: AI-DRIVEN REPAIR OF DISRUPTED
CAMPUS TRANSPORT SCHEDULES**

Submitted in partial fulfillment of the requirements

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ABSTRACT

Campus transportation schedules can become inefficient when unexpected disruptions such as bus breakdowns, road closures, or sudden increases in passenger demand occur. Route2Camp is an AI-driven system designed to repair existing campus transport schedules while minimizing passenger disruption and maintaining essential transport constraints. The system models campus roads, bus routes, stops, vehicle capacities, and passenger demand using a transport simulator. When a disruption occurs, a Reinforcement Learning (RL) agent analyses the current transport state and selects suitable repair actions, such as reassigning stops, reallocating passengers, modifying routes, or adjusting departure times. Gymnasium and PyTorch are used for the learning environment and model, while NetworkX represents the campus road network and supports alternative route selection. An OR-Tools-based constraint layer verifies vehicle capacity, road availability, and service requirements. A FastAPI-based dashboard displays the disruption and repaired schedule. The system is evaluated against greedy reassignment and optimization-based approaches using passenger waiting time, additional distance, unserved demand, and schedule changes. Route2Camp aims to provide an adaptive and efficient approach for recovering campus transport services with minimal changes to the original schedule.

Keywords Route2Camp, Artificial Intelligence, Reinforcement Learning, Campus Transportation, Schedule Repair, Route Optimization

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